

Pratik Hadawale

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PROFESSIONAL EXPERIENCE

Junior Software Developer, CAD+T Consulting GmbH

2023 Oct – present
Coimbatore, India

- **Optimized** 3D Furniture simulator, reducing draw calls from **7000+** to **7** per frame, network calls from **80+** to **<5**, and loading times from **7 mins** to **<8s**. Used geometry merging, custom **shaders**, mesh **batching**, and texture/material *sharing*, resulting in consistent **60 FPS**.
- **Implemented Path-Tracing** with custom workflow for realistic rendering, including integration in Webpack / Vite for build.
- **Optimized** 3D simulation engine **FPS** from **4** to **60**, reduced draw calls by **97%** (21,664 to 745), and decreased loading time from **62.3s** to **7.9s** through optimized geometry creation, cached/shared materials.
- **Developed** machining operations enabling cut-outs (e.g., CNC machines) and "mitter joint" functionality using **CSG**.
- **Implemented** dynamic stadium-like lighting, **HDR/shaders** integration, and shadow effects, enhancing rendering accuracy.
- **Fixed** critical bugs affecting 2D / 3D simulations, and **enhanced** UX through better visual cues.
- **Supported** and guided a **cross-functional** team while managing individual deadlines.
- **Refactored** the codebase, reducing its size by 40% while preserving all features and functionality.
- **Architected** an end to end 3D rendering pipeline for the core engine powering all simulations and a 3D Room Planner.

PROJECTS & OPEN SOURCE CONTRIBUTIONS

Technical Creative Projects Showcase, *Explore my projects with descriptions and visuals*

<https://github.com/irrevocablesake> 

Open Source Contributions

Link: <https://projects.blender.org/irrevocableSake> 

Learnt: Hands-on experience with Real World implementations of Vulkan, OpenGL, C++ and Python

Info: Recently began contributing to **Blender** to enhance skills and give back to the community

OpenGL, *Explore my journey with OpenGL*

Link: <https://github.com/irrevocablesake/OpenGL> 

Notable Projects:

▪ Quasar

Link: <https://github.com/irrevocablesake/Quasar> 

Learnt: PathTracing, BRDFs, BVH, Lighting, Textures, Geometry and Motion Blur

Info: Advanced CPU Ray Tracing Engine

▪ Grass Field Simulation (Procedural)

Link: <https://github.com/irrevocablesake/Grass-Field-Simulation> 

Learnt: Procedural Generation (shaders), LOD, Frustum Culling, Terrain Tiling

Info: A simulation involving upto 500k individual grass strands with wind mechanics.

SKILLS

3D Graphics Libraries / APIs

Three.js, OpenGL, GLSL, SFML, Blender, R3F, Path Tracing

Languages

Javascript, C++

ART STATION

2D & 3D ART

Works: https://www.artstation.com/pratikhadawale_sake 

Animation Project: <https://www.artstation.com/artwork/6NxeAV> 

AWARDS

Best Performer, CAD+T Solutions GmbH 2025

Rated as **Best Performer** in Annual Review

Felicitation with Gift, CAD+T Solutions GmbH 2024

- **Mentored professors** on JSReports as part of a company's initiative to bridge gap between university curriculum and industry standards

Trophy and Certificate, KBP College

- Got **trophy** and **certificate** for achieving 1st rank, in the First Year of Degree
- **Honorable** mention for achieving 2nd rank (in 3rd Year of Degree) during convocation

EDUCATION

Bachelors of Science in Computer Science, KBP College 2019 – 2022

CGPA for six semesters: 9.92 / 10

Navi Mumbai, India

Related Modules: DBMS, Disc Math, DSA, .Net, Web Dev, Comp Networks